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THERMAL MANAGEMENT SYSTEM For Electric Vehicles

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Recently some electric bikes caught fire due to failure of battery management system because of poor battery design. Unfortunately, proper thermal management is not there in most electric vehicles at present. So, here is a project to monitor the battery temperature and smoke detection to alert the EV user and stop the EV well in time to avoid further damage. The authors' prototype of the thermal management system is shown in Fig. 1 and its block diagram is shown in Fig. 2.

Lithium-ion battery has a specified operational temperature of -20°C to +60°C. It will provide the longest service life if its temperature is always kept within approximately 10°C to 40°C.

The proposed system, powered by a 12V battery, is built around

BILL OF MATERIAL

Components	Quantity	Description
ESP32 (U1)	1	ESP32 development doard
DS18B20 (U2)	1	Temperature sensor
MQ135 (U3)	1	Smoke detector
RG1602A (U4)	1	I2C 16x2 LCD display
LM7805 (U5)	1	5V voltage regulator
10k (R1)	1	Potmeter
1k (R2, R3)	2	¼-watt resistor
1N4009 (D1, D2)	2	Rectifier diode
100nF (C1, C2)	2	Ceramic disk capacitor
SU1	1	Buzzer
GU-SH112D (RL1)	1	High-power switching relay
BC547A (Q1, Q2)	2	Transistor (npn type)
On/off switch (S1)	1	SPDT switch
12V battery (BT1)	1	Lithium-ion battery
60V/40A/2.4kW (BT2)	1	Electric vehicle battery
Smartphone	1	Android mobile phone
Arduino Blue Control app		Need to download
IDE Arduino software		Need to download

Fig. 1: The authors' prototype

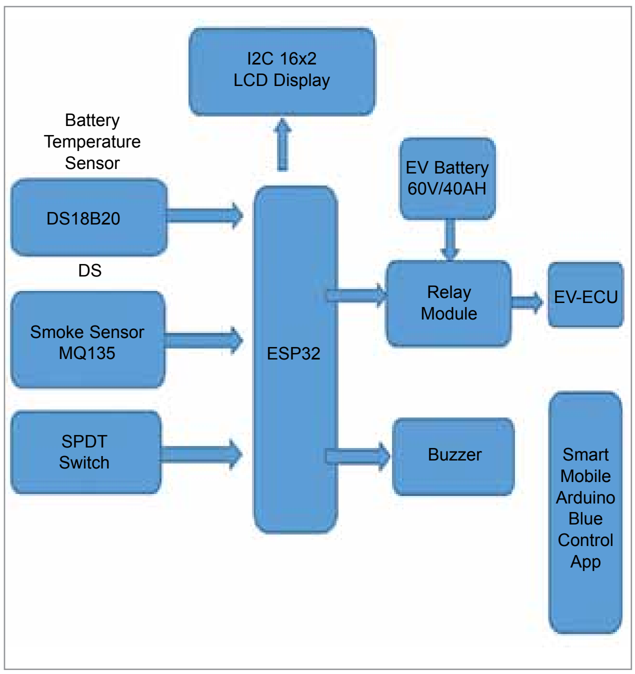

Fig. 2: Block diagram of thermal management system